

# BilenkinDSC680\_Week 2\_Credit Card Customer Churn Prediction

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## 1 Credit Card Customer Churn Prediction

### 1.1 1. Import Libraries

```
[167]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, roc_auc_score, confusion_matrix, \
    classification_report
```

### 1.2 2. Load Dataset

```
[168]: df = pd.read_csv("BankChurners.csv")
df.head()
```

```
[168]: CLIENTNUM      Attrition_Flag  Customer_Age  Gender  Dependent_count  \
0  768805383  Existing Customer           45      M              3
1  818770008  Existing Customer           49      F              5
2  713982108  Existing Customer           51      M              3
3  769911858  Existing Customer           40      F              4
4  709106358  Existing Customer           40      M              3

      Education_Level  Marital_Status  Income_Category  Card_Category  \
0      High School      Married      $60K - $80K      Blue
1      Graduate      Single  Less than $40K      Blue
2      Graduate      Married      $80K - $120K      Blue
3      High School      Unknown  Less than $40K      Blue
4      Uneducated      Married      $60K - $80K      Blue

      Months_on_book  ...  Credit_Limit  Total_Revolving_Bal  Avg_Open_To_Buy  \
0           39  ...      12691.0           777      11914.0
1           44  ...       8256.0           864       7392.0
```

|   |    |     |        |      |        |
|---|----|-----|--------|------|--------|
| 2 | 36 | ... | 3418.0 | 0    | 3418.0 |
| 3 | 34 | ... | 3313.0 | 2517 | 796.0  |
| 4 | 21 | ... | 4716.0 | 0    | 4716.0 |

|   | Total_Amt_Chng_Q4_Q1 | Total_Trans_Amt | Total_Trans_Ct | Total_Ct_Chng_Q4_Q1 | \ |
|---|----------------------|-----------------|----------------|---------------------|---|
| 0 | 1.335                | 1144            | 42             | 1.625               |   |
| 1 | 1.541                | 1291            | 33             | 3.714               |   |
| 2 | 2.594                | 1887            | 20             | 2.333               |   |
| 3 | 1.405                | 1171            | 20             | 2.333               |   |
| 4 | 2.175                | 816             | 28             | 2.500               |   |

|   | Avg_Utilization_Ratio | \ |
|---|-----------------------|---|
| 0 | 0.061                 |   |
| 1 | 0.105                 |   |
| 2 | 0.000                 |   |
| 3 | 0.760                 |   |
| 4 | 0.000                 |   |

|   | Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_Dependent_count_Education_Level_Months_Inactive_12_mon_1 | \ |
|---|------------------------------------------------------------------------------------------------------------------------------------|---|
| 0 | 0.000093                                                                                                                           |   |
| 1 | 0.000057                                                                                                                           |   |
| 2 | 0.000021                                                                                                                           |   |
| 3 | 0.000134                                                                                                                           |   |
| 4 | 0.000022                                                                                                                           |   |

|   | Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_Dependent_count_Education_Level_Months_Inactive_12_mon_2 |
|---|------------------------------------------------------------------------------------------------------------------------------------|
| 0 | 0.99991                                                                                                                            |
| 1 | 0.99994                                                                                                                            |
| 2 | 0.99998                                                                                                                            |
| 3 | 0.99987                                                                                                                            |
| 4 | 0.99998                                                                                                                            |

[5 rows x 23 columns]

### 1.3 3. Data Understanding

```
[169]: df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10127 entries, 0 to 10126
Data columns (total 23 columns):
 #   Column
Non-Null Count  Dtype
---  -
```

```

-----
0  CLIENTNUM
10127 non-null int64
1  Attrition_Flag
10127 non-null object
2  Customer_Age
10127 non-null int64
3  Gender
10127 non-null object
4  Dependent_count
10127 non-null int64
5  Education_Level
10127 non-null object
6  Marital_Status
10127 non-null object
7  Income_Category
10127 non-null object
8  Card_Category
10127 non-null object
9  Months_on_book
10127 non-null int64
10 Total_Relationship_Count
10127 non-null int64
11 Months_Inactive_12_mon
10127 non-null int64
12 Contacts_Count_12_mon
10127 non-null int64
13 Credit_Limit
10127 non-null float64
14 Total_Revolving_Bal
10127 non-null int64
15 Avg_Open_To_Buy
10127 non-null float64
16 Total_Amt_Chng_Q4_Q1
10127 non-null float64
17 Total_Trans_Amt
10127 non-null int64
18 Total_Trans_Ct
10127 non-null int64
19 Total_Ct_Chng_Q4_Q1
10127 non-null float64
20 Avg_Utilization_Ratio
10127 non-null float64
21 Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_D
ependent_count_Education_Level_Months_Inactive_12_mon_1 10127 non-null float64
22 Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_D
ependent_count_Education_Level_Months_Inactive_12_mon_2 10127 non-null float64
dtypes: float64(7), int64(10), object(6)

```

memory usage: 1.8+ MB

```
[169]:
```

|       | CLIENTNUM    | Customer_Age | Dependent_count | Months_on_book | \ |
|-------|--------------|--------------|-----------------|----------------|---|
| count | 1.012700e+04 | 10127.000000 | 10127.000000    | 10127.000000   |   |
| mean  | 7.391776e+08 | 46.325960    | 2.346203        | 35.928409      |   |
| std   | 3.690378e+07 | 8.016814     | 1.298908        | 7.986416       |   |
| min   | 7.080821e+08 | 26.000000    | 0.000000        | 13.000000      |   |
| 25%   | 7.130368e+08 | 41.000000    | 1.000000        | 31.000000      |   |
| 50%   | 7.179264e+08 | 46.000000    | 2.000000        | 36.000000      |   |
| 75%   | 7.731435e+08 | 52.000000    | 3.000000        | 40.000000      |   |
| max   | 8.283431e+08 | 73.000000    | 5.000000        | 56.000000      |   |

|       | Total_Relationship_Count | Months_Inactive_12_mon | \ |
|-------|--------------------------|------------------------|---|
| count | 10127.000000             | 10127.000000           |   |
| mean  | 3.812580                 | 2.341167               |   |
| std   | 1.554408                 | 1.010622               |   |
| min   | 1.000000                 | 0.000000               |   |
| 25%   | 3.000000                 | 2.000000               |   |
| 50%   | 4.000000                 | 2.000000               |   |
| 75%   | 5.000000                 | 3.000000               |   |
| max   | 6.000000                 | 6.000000               |   |

|       | Contacts_Count_12_mon | Credit_Limit | Total_Revolving_Bal | \ |
|-------|-----------------------|--------------|---------------------|---|
| count | 10127.000000          | 10127.000000 | 10127.000000        |   |
| mean  | 2.455317              | 8631.953698  | 1162.814061         |   |
| std   | 1.106225              | 9088.776650  | 814.987335          |   |
| min   | 0.000000              | 1438.300000  | 0.000000            |   |
| 25%   | 2.000000              | 2555.000000  | 359.000000          |   |
| 50%   | 2.000000              | 4549.000000  | 1276.000000         |   |
| 75%   | 3.000000              | 11067.500000 | 1784.000000         |   |
| max   | 6.000000              | 34516.000000 | 2517.000000         |   |

|       | Avg_Open_To_Buy | Total_Amt_Chng_Q4_Q1 | Total_Trans_Amt | Total_Trans_Ct | \ |
|-------|-----------------|----------------------|-----------------|----------------|---|
| count | 10127.000000    | 10127.000000         | 10127.000000    | 10127.000000   |   |
| mean  | 7469.139637     | 0.759941             | 4404.086304     | 64.858695      |   |
| std   | 9090.685324     | 0.219207             | 3397.129254     | 23.472570      |   |
| min   | 3.000000        | 0.000000             | 510.000000      | 10.000000      |   |
| 25%   | 1324.500000     | 0.631000             | 2155.500000     | 45.000000      |   |
| 50%   | 3474.000000     | 0.736000             | 3899.000000     | 67.000000      |   |
| 75%   | 9859.000000     | 0.859000             | 4741.000000     | 81.000000      |   |
| max   | 34516.000000    | 3.397000             | 18484.000000    | 139.000000     |   |

|       | Total_Ct_Chng_Q4_Q1 | Avg_Utilization_Ratio | \ |
|-------|---------------------|-----------------------|---|
| count | 10127.000000        | 10127.000000          |   |
| mean  | 0.712222            | 0.274894              |   |
| std   | 0.238086            | 0.275691              |   |
| min   | 0.000000            | 0.000000              |   |

|     |          |          |
|-----|----------|----------|
| 25% | 0.582000 | 0.023000 |
| 50% | 0.702000 | 0.176000 |
| 75% | 0.818000 | 0.503000 |
| max | 3.714000 | 0.999000 |

| Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon |              |
|---------------------------------------------------------------------------|--------------|
| _Dependent_count_Education_Level_Months_Inactive_12_mon_1 \               |              |
| count                                                                     | 10127.000000 |
| mean                                                                      | 0.159997     |
| std                                                                       | 0.365301     |
| min                                                                       | 0.000008     |
| 25%                                                                       | 0.000099     |
| 50%                                                                       | 0.000181     |
| 75%                                                                       | 0.000337     |
| max                                                                       | 0.999580     |

| Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon |              |
|---------------------------------------------------------------------------|--------------|
| _Dependent_count_Education_Level_Months_Inactive_12_mon_2                 |              |
| count                                                                     | 10127.000000 |
| mean                                                                      | 0.840003     |
| std                                                                       | 0.365301     |
| min                                                                       | 0.000420     |
| 25%                                                                       | 0.999660     |
| 50%                                                                       | 0.999820     |
| 75%                                                                       | 0.999900     |
| max                                                                       | 0.999990     |

The dataset contains 10,127 observations and 23 variables. It includes a mix of categorical and numerical features. No missing values were found in the dataset, indicating good data quality.

## 1.4 4. Data Cleaning

```
[170]: df = df.drop([
    ↵
    ↵ 'Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_Dependent_count_
    ↵
    ↵ 'Naive_Bayes_Classifier_Attrition_Flag_Card_Category_Contacts_Count_12_mon_Dependent_count_
], axis=1)
```

### Convert Target Variable

```
[171]: df['Attrition_Flag'] = df['Attrition_Flag'].map({
    'Existing Customer': 0,
    'Attrited Customer': 1
})
```

### Check Missing Values

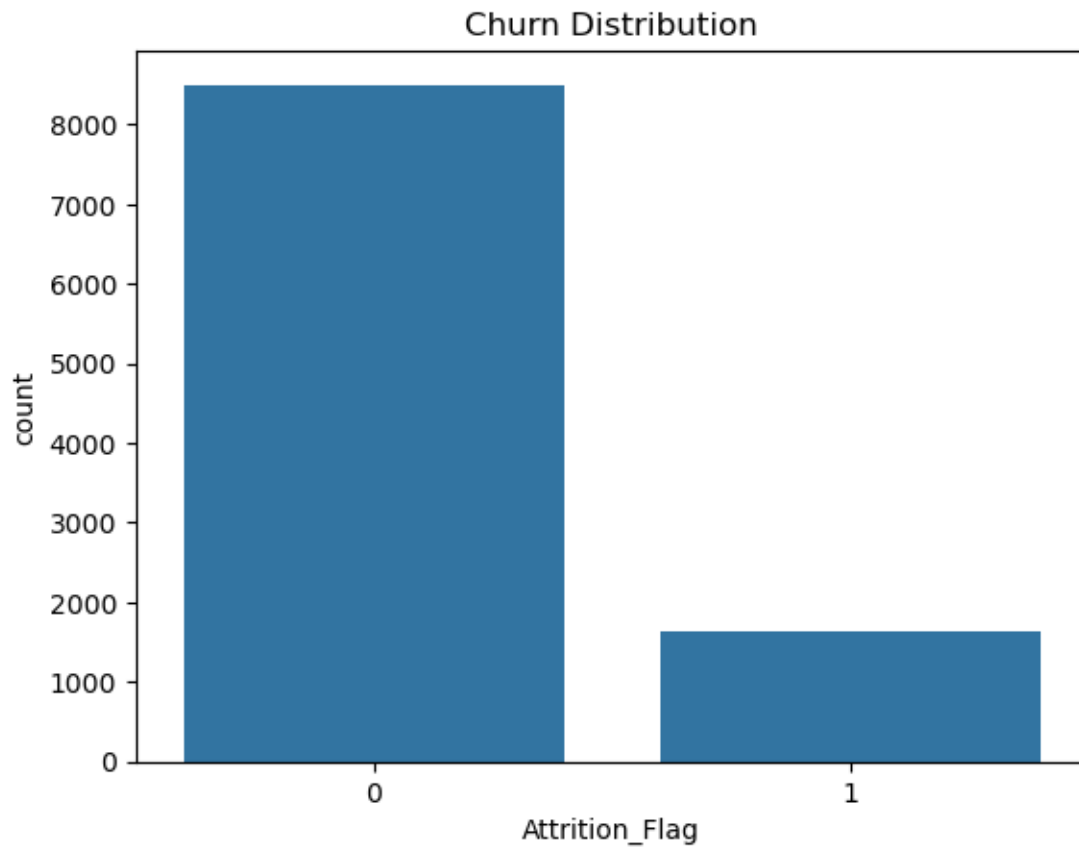
```
[172]: df.isnull().sum()
```

```
[172]: CLIENTNUM          0
      Attrition_Flag      0
      Customer_Age        0
      Gender              0
      Dependent_count     0
      Education_Level     0
      Marital_Status      0
      Income_Category     0
      Card_Category       0
      Months_on_book      0
      Total_Relationship_Count 0
      Months_Inactive_12_mon 0
      Contacts_Count_12_mon 0
      Credit_Limit        0
      Total_Revolving_Bal  0
      Avg_Open_To_Buy     0
      Total_Amt_Chng_Q4_Q1 0
      Total_Trans_Amt      0
      Total_Trans_Ct       0
      Total_Ct_Chng_Q4_Q1  0
      Avg_Utilization_Ratio 0
      dtype: int64
```

## 1.5 5. Exploratory Data Analysis (EDA)

### 1. Churn Distribution

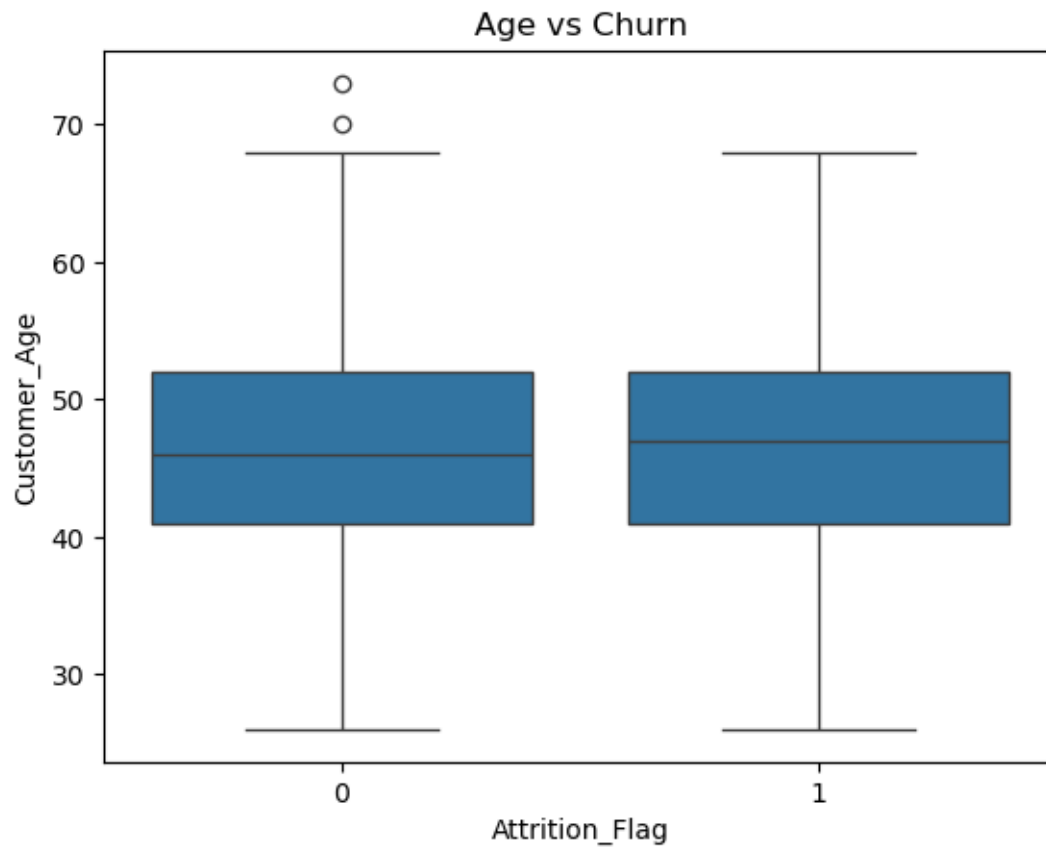
```
[173]: sns.countplot(x='Attrition_Flag', data=df)
      plt.title("Churn Distribution")
      plt.savefig("Churn_Distribution", bbox_inches='tight')
      plt.show()
```



Note: 0 represents existing customers and 1 represents customers who have churned.

#### Age vs Churn

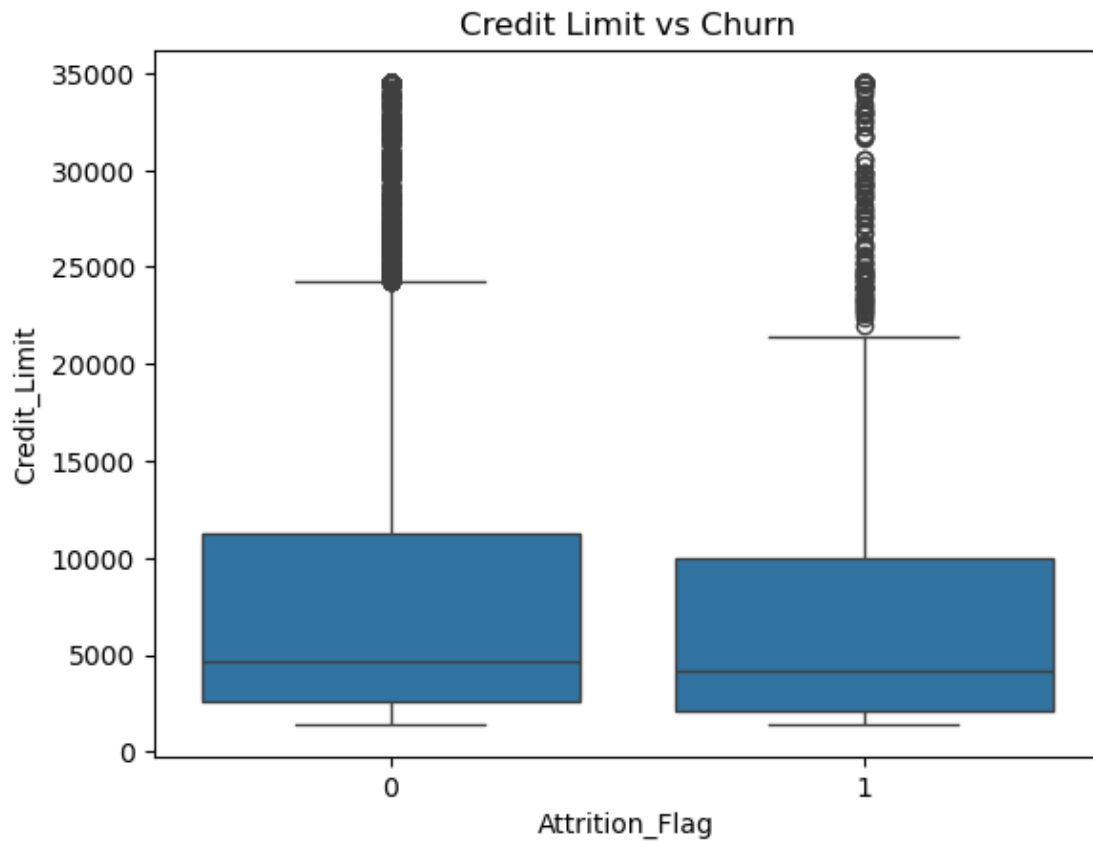
```
[174]: sns.boxplot(x='Attrition_Flag', y='Customer_Age', data=df)
plt.title("Age vs Churn")
plt.savefig("Age_vs_Churn", bbox_inches='tight')
plt.show()
```



#### Credit Limit vs Churn

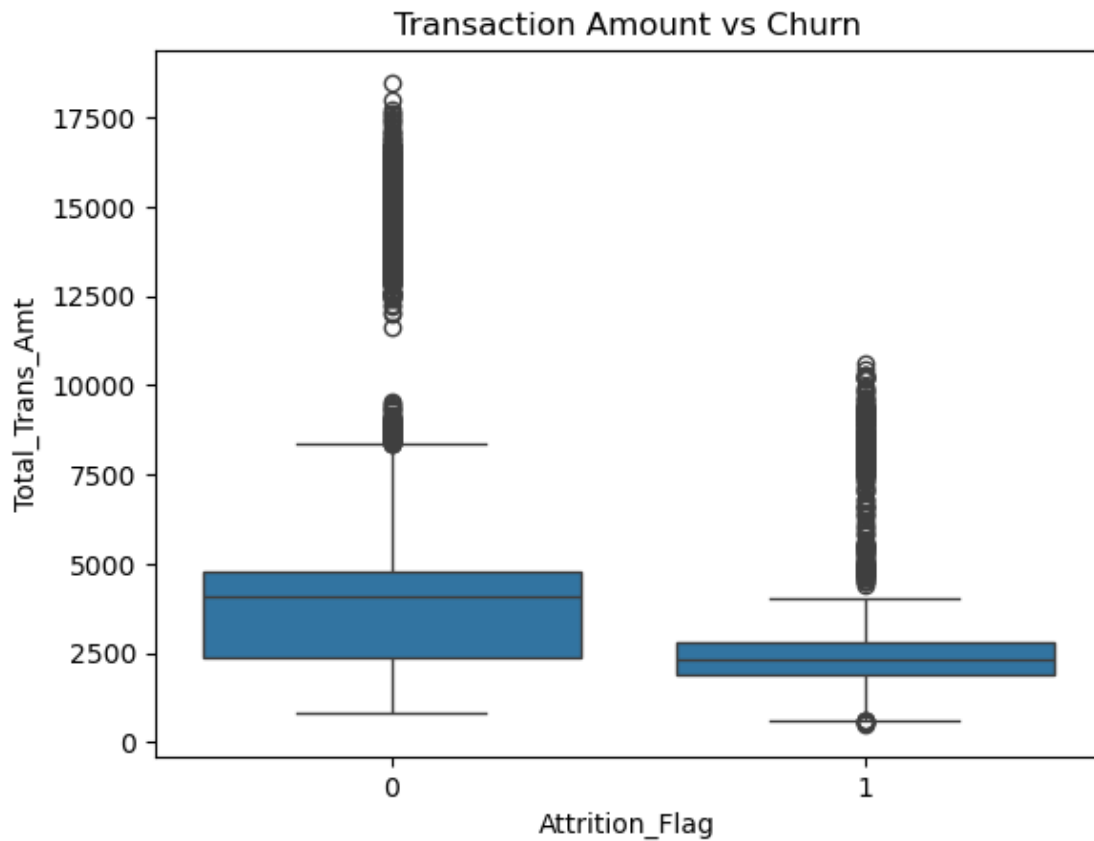
```
[175]: sns.boxplot(x='Attrition_Flag', y='Credit_Limit', data=df)
plt.title("Credit Limit vs Churn")
plt.savefig("Credit_Limit_vs_Chur", bbox_inches='tight')
plt.show()
```





#### Transactions vs Churn

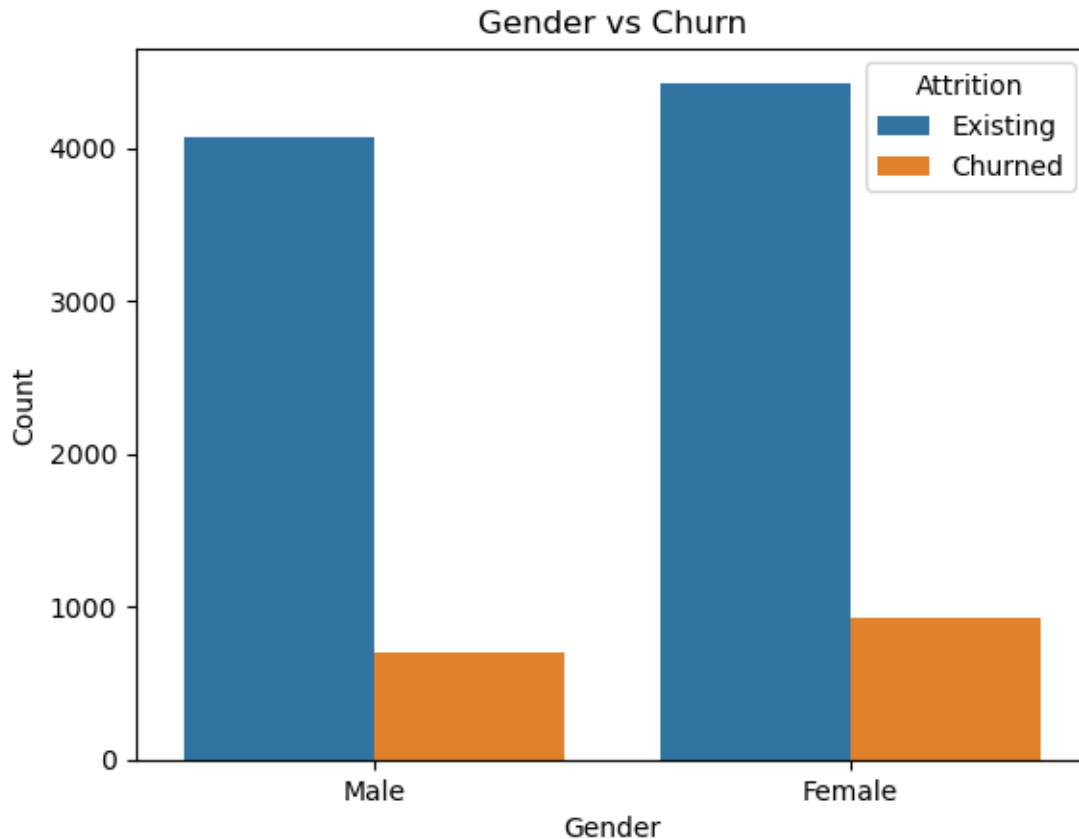
```
[176]: sns.boxplot(x='Attrition_Flag', y='Total_Trans_Amt', data=df)
plt.title("Transaction Amount vs Churn")
plt.savefig("Transaction_Amount_vs_Churn", bbox_inches='tight')
plt.show()
```



#### Gender vs Churn

```
[177]: df['Gender_Label'] = df['Gender'].map({'M': 'Male', 'F': 'Female'})

sns.countplot(x='Gender_Label', hue='Attrition_Flag', data=df)
plt.title("Gender vs Churn")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.legend(title="Attrition", labels=["Existing", "Churned"])
plt.savefig("Gender_vs_Churn.png", bbox_inches='tight')
plt.show()
```



The chart shows the distribution of customers by gender and churn status. Both male and female customers follow a similar pattern, indicating that gender does not significantly impact churn behavior.

## 1.6 6. Feature Engineering

### Encode Categorical Variables

```
[178]: categorical_cols = df.select_dtypes(include='object').columns

le = LabelEncoder()
for col in categorical_cols:
    df[col] = le.fit_transform(df[col])
```

## 1.7 7. Modeling

### Split Data

```
[179]: X = df.drop('Attrition_Flag', axis=1)
y = df['Attrition_Flag']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
↪random_state=42)
```

### Logistic Regression

```
[180]: lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
```

### Random Forest

```
[181]: rf = RandomForestClassifier()
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)
```

## 1.8 8. Model Evaluation

```
[182]: print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))

print("ROC-AUC (RF):", roc_auc_score(y_test, rf.predict_proba(X_test)[:,:1]))
```

Logistic Regression Accuracy: 0.8611385324119776

Random Forest Accuracy: 0.953932214544258

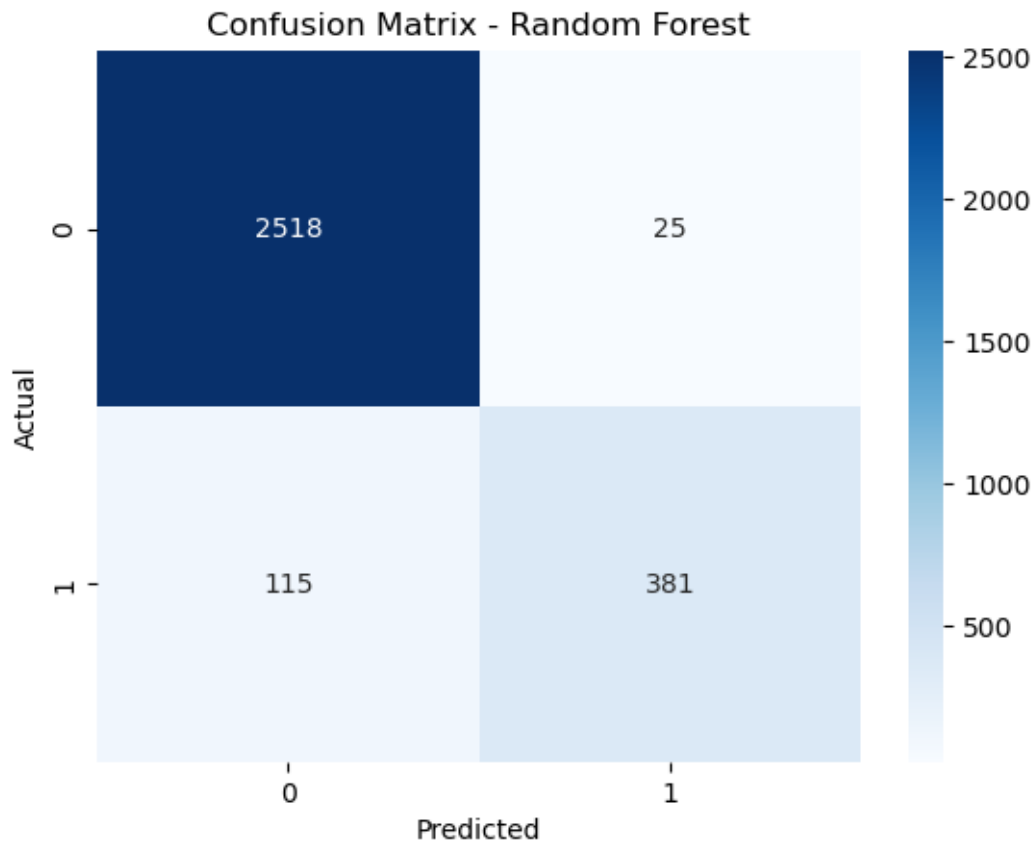
ROC-AUC (RF): 0.9870715626704553

### Confusion Matrix

```
[183]: import seaborn as sns

cm = confusion_matrix(y_test, y_pred_rf)

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Random Forest")
plt.show()
```



## 1.9 9. Feature Importance

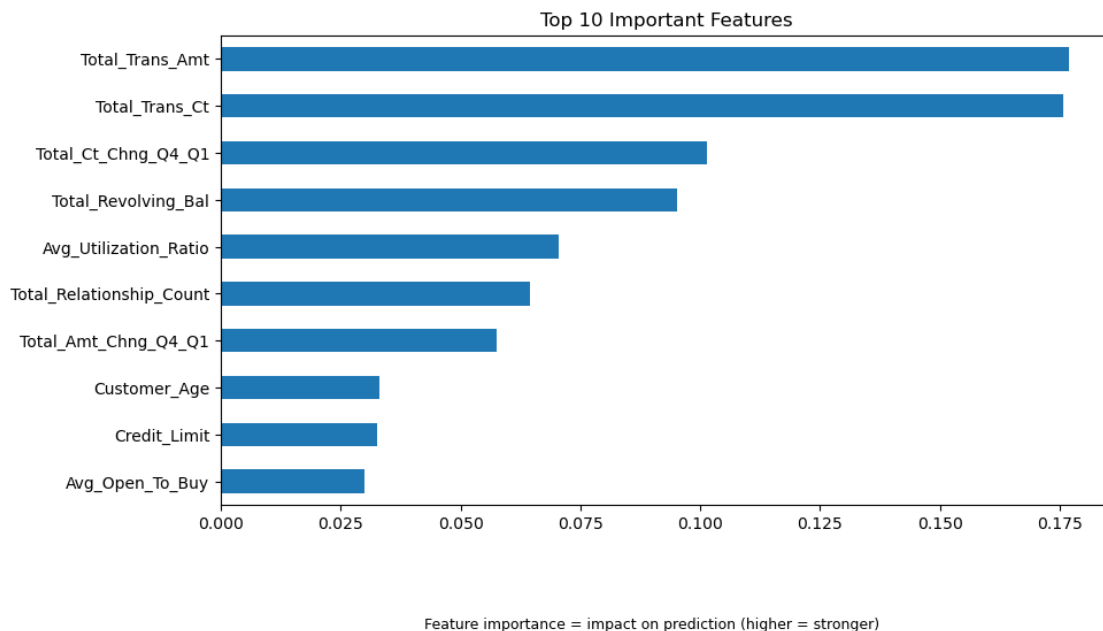
```
[184]: importances = rf.feature_importances_  
features = X.columns  
  
feat_importance = pd.Series(importances, index=features).  
    ↪ sort_values(ascending=False)  
  
# Keep only top 10  
top10 = feat_importance.head(10)  
  
plt.figure(figsize=(10,6))  
  
# Horizontal bar chart  
top10.sort_values().plot(kind='barh')  
  
plt.title("Top 10 Important Features")  
  
# Add space at bottom
```

```
plt.subplots_adjust(bottom=0.2)

# Add note below chart
plt.figtext(0.5, 0.02,
"Feature importance = impact on prediction (higher = stronger)",
ha="center", fontsize=9)

plt.savefig("Top_10_Important_Features.png", bbox_inches='tight')

plt.show()
```



## 1.10 10. Conclusion

Based on the data analysis, the dataset appears to be clean and of good quality. The two Naive Bayes classifier columns were removed because they are not actual predictive features and could introduce data leakage. The dataset does not contain any missing values, which simplifies the preprocessing step.

From the churn distribution, the dataset is imbalanced, with approximately 8,500 existing customers and around 1,500 customers who have churned. This indicates that most customers remain with the bank, while a smaller portion leave.

Looking at the Age vs Churn analysis, both groups have a very similar age distribution. The average age is around 46 years for both existing and churned customers. The presence of a few outliers does not significantly affect the overall distribution. This suggests that age is not a strong factor influencing customer churn.

Similarly, the Credit Limit vs Churn analysis shows that both groups have comparable distributions,

with many outliers present in both categories. The median values are also very close. This indicates that credit limit alone is not a decisive factor in whether a customer stays or leaves.

However, the Total Transaction Amount shows a clear difference between the two groups. Customers who churn tend to have significantly lower transaction amounts compared to those who remain. This suggests that lower engagement with the credit card may be a strong indicator of potential churn. Customers with lower spending activity are more likely to leave, which can be useful for identifying at-risk customers.

The Gender vs Churn analysis shows no significant difference between male and female customers. Both groups display a similar distribution of existing and churned customers, suggesting that gender does not play a major role in predicting customer churn.

From the modeling results, Logistic Regression achieved an accuracy of approximately 85.9%, while Random Forest performed significantly better with an accuracy of about 96.0% and a ROC-AUC score of 0.988. This indicates that the Random Forest model is highly effective in distinguishing between churned and existing customers.

The confusion matrix shows that the model correctly classified the majority of customers, with a relatively small number of misclassifications. This further confirms that the model performs well.

Finally, the feature importance analysis shows that Total Transaction Amount, Total Transaction Count, and Total Count Change are among the most important factors in predicting churn. These features are directly related to customer activity and engagement, which supports the earlier findings from the EDA.

Overall, the results suggest that customer behavior and transaction activity are key drivers of churn, while demographic features such as age and gender have less impact.